# Sample Inherent Correlation Mining for Hyperspectral Image Classification

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***Abstract*—Graph convolutional networks (GCNs) have recently shown great promise in hyperspectral image classification (HIC) by effectively modeling long-range correlations among samples. However, existing methods often rely on sample similarity for graph construction while neglecting intra-class variability, thereby limiting the robustness of the extracted features. This article introduces a novel HIC method that leverages sample inherent correlations to enhance feature representation and improve classification performance. Specifically, for the labeled sample set, two distinct intra-class sample graphs are constructed: the intra-class sample similarity graph (ISSG) for capturing class cohesiveness, and the intra-class sample discrepancy graph (ISDG) for modeling variability within the same class. These complementary graph structures are integrated into a dual-branch GCN model, where the ISSG-based branch acts as the primary classifier and the ISDG-based branch serves as an auxiliary component. During training, contrastive learning is employed to enhance feature alignment and generalization. For testing, category prediction is performed on the primary branch using a graph structure constructed from an augmented sample set, comprising training samples, test samples, and the spatial neighbors of the test samples within a predefined local window based on similarity metrics. Finally, the class label for each test sample is determined as the dominant semantic category within its local spatial window. Experimental results on four public hyperspectral image (HIS) datasets demonstrate that the proposed method outperforms state-of-the-art techniques and existing graph-based approaches. The code will be available at https://github.com/fengjia116/SICM**

***Index Terms*—Contrastive learning, graph convolutional network (GCN), hyperspectral image classification (HIC), sample inherent correlation.**

## I. Introduction

HYPERSPECTRAL images (HSIs) provide unique spectral signatures across a broad spectrum range with fine spectral resolution, together with spatial information that reflects the spatial distribution and structural characteristics of various ground objects [1], [2], [3]. The rich spectral-spatial information inherent in HSIs supports a wide range of applications, such as precision agriculture [4], urban planning [5], and environmental monitoring [6]. Accordingly, hyperspectral image classification (HIC) has been extensively studied as a fundamental tool for these practical applications [7]. However, the high dimensionality and complex spatial structures of HSIs pose significant challenges to image analysis and landcover classification. Achieving accurate per-pixel classification of HSIs remains a formidable task in real-world applications.

Considering that the high spectral correlation between adjacent bands introduces significant redundancy and crosstalk [8], early HIC methods primarily focused on reducing information redundancy and extracting spectral features using techniques, such as principal component analysis (PCA) [9], neighborhood preserving embedding (NPE) [10], and local Fisher's discriminant analysis (LFDA) [11]. However, spectral information is often sensitive to environmental factors, such as atmospheric conditions and illumination. Consequently, these methods may suffer from misclassification because of large spectral variability within the same land-cover class and similar spectral characteristics across different categories of land covers.

Since ground objects often exhibit distinct spatial patterns, a series of feature extraction methods has been developed to more fully exploit spectral-spatial information, reduce uncertainty in spectral data, and enhance inter-class separability [12]. Traditional methods extracted spectral and spatial features independently and then integrated them through feature concatenation or kernel functions for category identification [13]. Subsequent advancements introduced machine learning methods, such as manifold learning [14], sparse representation [15], and collaborative representation [16], to improve feature discriminability and robustness, thereby enhancing classification accuracy. Given that HSI data naturally exhibit a 3-D cube structure [17], [18], tensor-based approaches have been proposed to preserve spectral-spatial correlations for classification [19]. However, these approaches often rely on manually designed features, which limits their adaptability to the complex spectral-spatial relationships inherent in hyperspectral data.

The advent of deep learning has marked a significant breakthrough in HIC by simulating the neural structures of the human brain to enable hierarchical learning from low-level to high-level feature representations. Numerous convolutional neural network (CNN)-based methods have been developed for HIC, facilitating automated feature extraction and more

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effective integration of spectral-spatial information [20]. For instance, Hamida et al. [21] employed a 3-D CNN framework to preserve the intrinsic properties of the 3-D structure of hyperspectral data. Building on this foundation, recent studies have proposed strategies to further enhance the classification performance of CNN-based methods, such as multiscale feature extraction schemes, advanced convolution operations, and attention mechanisms. For example, Zhu et al. [22] introduced a multiscale feature fusion network to integrate features from different scales, improving classification accuracy. Yang et al. [23] proposed a multiscale feature aggregation network that adaptively fuses features from different scales to capture both local details and global context. Li et al. [24] developed a lightweight CNN with dynamic kernel generation, adapting to the unique characteristics of different spectral bands. Gao et al. [25] presented a dual-branch attention network that simultaneously processes spatial and spectral information, effectively capturing the intrinsic characteristics of HSIs.

More recently, neural architectures with stronger long-range dependency modeling capabilities, such as Transformers and state-space models, have driven significant advances in remote sensing image interpretation, including super-resolution [26], [27], denoising [28], anomaly detection [29], [30], change detection [31], and classification [32]. In the context of HIC, Zhong et al. [33] developed a spectral-spatial transformer network (SSTN) using a factorized architecture search framework to effectively capture both spectral and spatial information while optimizing the network structure. Chen et al. [34] designed a global spectral projection and space aggregation (GSPSA) approach to enhance spectral consistency within local neighborhoods. Liao et al. [35] proposed the spectral-spatial fusion transformer network ($S^2$FTNet), which combines SpaFormer and SpeFormer modules with an adaptive fusion mechanism to effectively integrate spatial and spectral features. In addition, Wang et al. [36] developed a spatial-spectral state space model ($S^2$Mamba), which leverages selective structured state-space models with linear complexity to improve classification performance. Han et al. [37] proposed a subpixel spectral variability network ($S^2$VNet) that combines a CNN-Transformer architecture with nonlinear unmixing to model spectral variability. Xu et al. [38] proposed the dual selective fusion transformer (DSFormer) network, which adaptively fuses multiscale spatial-spectral features through kernel selection and selectively retains relevant tokens during self-attention.

Although the aforementioned methods have significantly improved classification accuracy by leveraging spectral-spatial information and end-to-end feature learning, they still exhibit notable limitations. CNN-based methods often struggle to capture long-range dependencies and global contextual information, whereas transformer-based methods, although effective at modeling global relationships, are computationally expensive and may not fully exploit local spatial information. Moreover, these approaches generally fail to effectively model the non-Euclidean topological structure in the HSIs, which is crucial for accurate classification. Additionally, their complex frameworks often require large-scale labeled training samples for model optimization, making them prone to overfitting, particularly in resource-constrained scenarios.

Graph convolutional networks (GCNs) have emerged as a powerful paradigm for processing unstructured data and have attracted increasing attention in the field of HIC [39]. By employing message-passing mechanisms, GCNs can effectively model the non-Euclidean structure of HSI data and capture correlations from local to global scales, thereby enabling robust feature representation and improved classification performance [40]. Hong et al. [41] introduced a trainable end-to-end mini-GCN framework that combines the strengths of CNNs and GCNs to capture both local and global relationships in HSI data. Building on this foundation, numerous innovative GCN-based approaches have been proposed to enhance landcover classification performance. For instance, Wan et al. [42] presented a multiscale feature search-based GCN that adaptively selects optimal features at different scales, thereby addressing the challenges posed by varying spatial resolutions in HSIs. Qin et al. [43] developed an efficient graph convolutional random vector functional link network, improving both efficiency and accuracy. Yang et al. [44] introduced a GCN-Transformer fusion network that integrates the capabilities of GCNs and transformers, leveraging complementary advantages to enhance HIC performance.

In related work, Wang et al. [45] designed a double-branch framework incorporating multilevel skip connections and sparse attention mechanisms to enhance discriminative feature representation. Similarly, Cao et al. [46] developed a global-local manifold embedding broad GCN that integrates edge feature fusion with global-local manifold embedding. Yu et al. [47] proposed a discriminant superpixel graph that optimizes both within-superpixel and between-superpixel connections, reducing inter-class weights while enhancing intra-class connectivity. Jiang et al. [48] introduced a GraphGST framework that employs absolute positional encoding to capture positional relationships between pixels and cross-view contrastive learning to improve representation learning. Wang et al. [49] presented an enhanced hypergraph neural network for mining high-order relationships in hyperspectral data. Zhang et al. [50] proposed a non-Euclidean spectral-spatial feature mining network with a gated GCN-CNN (NESSGGCN) architecture, which combines non-Euclidean feature extraction with pixel-level CNNs through weighted fusion. In addition, Wang et al. [51] developed a weak-strong graph contrastive learning (WSGraphCL) framework for HIC, which constructs $K$-hop subgraphs with spectral-spatial adjacency matrices and employs contrastive learning to extract discriminative features.

Although GCN-based methods have achieved remarkable progress, they still fail to fully exploit available prior information. As illustrated in Fig. 1, an analysis of the University of Houston dataset reveals that, under Euclidean distance-based metrics, certain samples belonging to class label 1 (Healthy grass) often exhibit higher spectral similarity to samples from class label 2 than to other samples within their own category. This phenomenon, in which inter-class similarity exceeds intra-class similarity, poses significant challenges for

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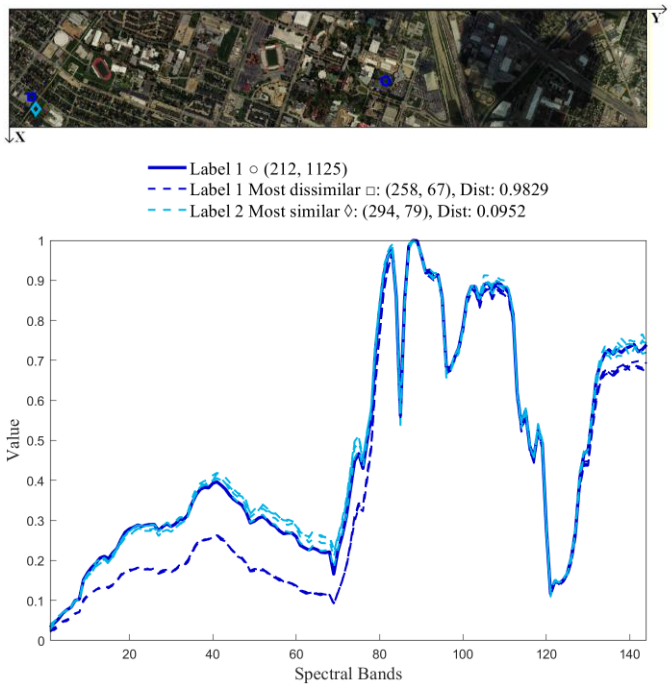



Fig. 1. Spectral response curves of different types of landcovers.

accurately identifying ground object categories. Existing methods have not conducted in-depth analyses of intra-class variability, nor have they effectively utilized prior knowledge, such as semantic information and spectral-spatial characteristics, to mine the inherent correlations among samples. These shortcomings constrain the robustness and discriminative power of the extracted features, resulting in unsatisfactory classification accuracy and limited generalization performance.

## II. Motivation and Contribution

Although GCN-based methods have significantly advanced HIC, several challenges and limitations persist. First, graph structures in existing methods are typically constructed using predefined similarity metrics based on spectral information. Such strategies mainly emphasize sample similarity while neglecting inherent intra-class diversity. In complex scenes, phenomena such as "different objects with similar spectra" and "the same object with different spectra" are particularly pronounced, making traditional distance metrics inadequate for accurately characterizing sample correlations. Second, the comprehensive exploitation of both local and global information is critical for accurate land-cover classification in HSIs. However, most existing GCN-based methods rely on auxiliary modules, such as CNNs or Transformers, to mine land-cover information. Although these approaches improve classification performance, they also substantially increase model complexity and generally require larger labeled training sets. Furthermore, prior information within a given scene is often underutilized, leading to suboptimal feature representations and unsatisfactory classification results. Despite notable advances, current methods still struggle to effectively capture the subtle intra-class and inter-class correlations inherent in hyperspectral data. Existing graph construction paradigms predominantly emphasize similarity-based connections while overlooking the discriminative information embedded in sample differences, particularly intra-class variations, which could provide complementary cues for classification.

Motivated by these challenges and recent advances, this article introduces a novel HIC method based on sample inherent correlation mining (SICM), which comprehensively leverages semantic information and spectral-spatial information to capture both local and global dependencies for robust feature extraction and accurate landcover classification. First, we integrate the spatial information derived from extended multiattribute profiles (EMAPs) with spectral bands to construct a comprehensive spectral-spatial representation of the original samples. To fully exploit the prior information available within a given scene, we then construct two distinct intra-class sample graphs: 1) the intra-class sample similarity graph (ISSG), which emphasizes consistency and cohesiveness within the same class, and 2) the intra-class sample discrepancy graph (ISDG), which is designed to capture intra-class sample variability . Subsequently, the spatial neighbors of each unknown test sample within a fixed-size spatial window, together with the labeled samples, are added to the original test sample set to form the augmented sample set of interest. Since the semantic category information of the unknown test samples is unavailable, the graph structure for the augmented test sample set is constructed based on spectral-spatial information similarity metrics. In order to comprehensively exploit the two graph structures, we propose a dual-branch GCN model integrated with contrastive learning for cross-branch information interaction. In this architecture, the ISSG-based branch acts as the primary branch, whereas the ISDG-based branch serves as the auxiliary branch. After training, the graph structure of the augmented test sample set, constructed according to high sample similarity, is employed by the primary branch to generate the initial prediction results. Finally, considering that samples belonging to the same land-cover class tend to be densely and continuously distributed within local spatial regions, each unknown test sample is assigned the dominant class label within its local spatial window. The main contributions of this article are summarized as follows.

1) A dual-perspective intra-class relationship mining framework is proposed to simultaneously capture intra-class sample similarity and variability, thereby enabling a more comprehensive characterization of complex sample relationships in hyperspectral data.
2) A dual-branch GCN model with a cross-branch optimization strategy is designed to jointly incorporate classification and contrastive learning objectives, thereby enhancing model generalization through the explicit modeling of both intra-class commonalities and variations.
3) A hierarchical information fusion strategy is introduced to progressively integrate multiple types of information, ensuring that relevant land-cover information is fully exploited at each processing stage and thereby improving the model's adaptability across diverse scenarios.

The remainder of this article is organized as follows. Section III provides a detailed description of the proposed method.

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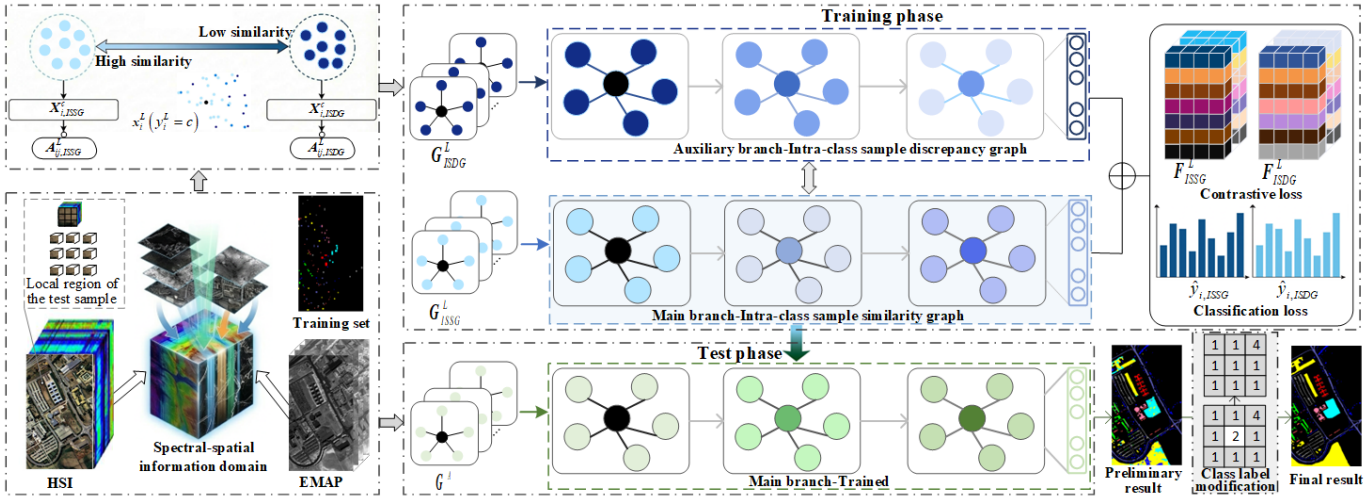



Fig. 2. Framework of the proposed method.

Section IV presents the experimental results and analysis. Section V concludes the work.

## III. Proposed Methodology

Fig. 2 illustrates the flowchart of the proposed method, which comprises three main components: 1) a separated graph construction scheme; 2) the dual-branch GCN model; and 3) a class label refinement strategy. Each component is described in detail as follows.

### A. Separated Graph Construction Scheme

Considering that EMAPs effectively capture diverse spatial characteristics of different structures in HSIs across multiple scales, we combine the spatial features extracted by EMAPs with the original hyperspectral data to construct sample representations in the spectral-spatial information domain [52]. Specifically, EMAP first applies PCA to reduce information redundancy and then constructs attribute profiles by performing a series of morphological operations with attribute filters (e.g., area, diagonal length, standard deviation, and moment of inertia) at multiple thresholds on each principal component. The resulting multiscale spatial features are concatenated with the original spectral bands to form a joint spectral-spatial representation [53], [54]. Formally, given the reconstructed hyperspectral data $\chi \in \mathbb{R}^{H\times W\times d_{SS}}$ where $H$ and $W$ represent the spatial height and width, respectively, and $d_{SS}$ denotes the dimension of the spectral-spatial information. To facilitate graph-based learning, let $\boldsymbol{X}^L \in \mathbb{R}^{N^L\times d_{SS}}$ denote the labeled sample set containing $N^L$ labeled samples. The corresponding label matrix of $\boldsymbol{X}^L$ is denoted by $\boldsymbol{Y}^L \in \mathbb{R}^{N^L\times C}$, where $C$ is the total number of class labels and the element $y^L_{i,c}$ in the $i$th row and $c$th column equals 1 if $\boldsymbol{x}^L_i$ belongs to class $c$, and 0 otherwise. Here, the objective is to predict the semantic categories of the unlabeled test sample set $\boldsymbol{X}^U \in \mathbb{R}^{N^U\times d_{SS}}$. Therefore, the joint use of spectral information and EMAP-derived spatial information establishes the foundation for sample representation and lays the groundwork for similarity measurement and relationship mining among samples. Unlike traditional GCN-based methods, which construct a single graph structure using both training and test samples based on sample similarity, this article proposes a separate graph construction scheme to fully exploit the available information.

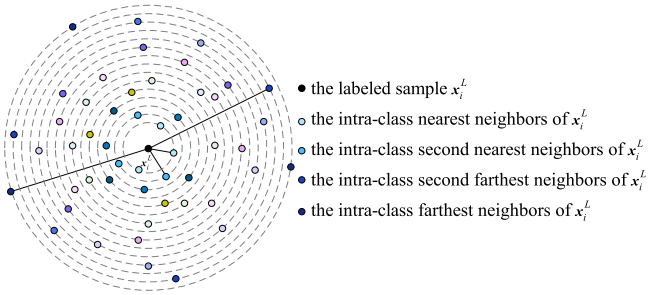



Fig. 3. Schematic of distance distribution between intra-class samples and the given sample.

For the labeled sample set, we use only labeled samples to construct two complementary graph structures: 1) an intra-class similarity-prioritized graph, termed ISSG and 2) an intra-class variability-prioritized graph, termed ISDG. The ISSG connects highly similar samples within the same class based on spectral-spatial distance metrics to capture class commonalities, whereas the ISDG links samples with greater intra-class differences to characterize the diversity and variability within a class. These complementary graph structures enable the model to comprehensively characterize both the consistency and variability within each semantic category. As shown in Fig. 3, for the labeled sample $\boldsymbol{x}^L_i$ belonging to a known semantic category $c$, the distance distribution between the sample and its intra-class counterparts is described based on Euclidean distance metrics derived from the re-expressed dataset in the spectral-spatial domain. Accordingly, the neighboring sample set $\boldsymbol{X}^c_{i,ISSG} \in \mathbb{R}^{k\times d_{SS}}$ comprised of the first $k$-nearest neighbors of $\boldsymbol{x}^L_i$ within the same class based on Euclidean distance [53], [55]. Similarly, let the sample set $\boldsymbol{X}^c_{i,ISDG} \in \mathbb{R}^{k\times d_{SS}}$ denote the first $k$ farthest neighbors of $\boldsymbol{x}^L_i$, indicating that the Euclidean distance between $\boldsymbol{x}^L_i$ and

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the sample $\boldsymbol{X}^c_{ik,ISDG}$ is $k$th farthest with respect to $\boldsymbol{x}^L_i$. In other words, the intra-class neighboring samples are selected according to two distinct criteria: samples with high intra-class spectral-spatial similarity are prioritized for $\boldsymbol{X}^c_{i,ISSG}$, whereas samples with greater intra-class spectral-spatial dissimilarity are selected for $\boldsymbol{X}^c_{i,ISDG}$.

Inspired by the attention coefficient mechanism in graph attention networks [56], [57], we introduce a novel weight modulation mechanism for intra-class samples that adaptively adjusts the edge weights of intra-class sample connections. Based on the preselected intra-class sample subsets $\boldsymbol{X}^c_{i,ISSG}$ and $\boldsymbol{X}^c_{i,ISDG}$, the edge weights between different nodes in the adjacency matrices $\boldsymbol{A}^L_{ISSG}$ and $\boldsymbol{A}^L_{ISDG}$ for ISSG and ISDG, respectively, are computed using the radial basis function (RBF). Formally, the corresponding elements in the adjacency matrices $\boldsymbol{A}^L_{ISSG}$ and $\boldsymbol{A}^L_{ISDG}$ are defined as follows:

$$\begin{aligned}\boldsymbol{A}^L_{ij,ISSG} &= \begin{cases}\exp\left(-\left\|\boldsymbol{x}^L_i-\boldsymbol{x}^L_j\right\|^2/\sigma^2\right), & \text{if } y^L_i=y^L_j\\ \qquad\qquad = c \text{ and } \boldsymbol{x}^L_j\in\boldsymbol{X}^c_{i,ISSG} & \\ 0, & \text{else}\end{cases}\\ &= \begin{cases}\dfrac{1}{1+\gamma\exp\left(-\left\|\boldsymbol{x}^L_i-\boldsymbol{x}^L_j\right\|^2/\sigma^2\right)}, & \text{if } y^L_i=y^L_j\\ \qquad\qquad = c \text{ and } \boldsymbol{x}^L_j\in\boldsymbol{X}^c_{i,ISDG} & \\ 0, & \text{else}\end{cases}\end{aligned} \tag{1}$$

where $\delta$ represents the width of the RBF, which controls the sensitivity to spectral-spatial distance, and $\gamma$ is a hyperparameter, termed the intra-class variability modulation coefficient, constrained to the range [0, 1]. The parameter $\gamma$ governs the weight distribution among intra-class heterogeneous samples through an inverse-distance weighting mechanism, enabling the model to assign adaptive weights to highly variable sample pairs within the same class. As $\gamma$ decreases, the contribution of the most dissimilar intra-class samples is amplified, whereas the influence of moderately dissimilar samples is diminished.

Unlike conventional similarity-based graphs, which primarily model intra-class cohesion, the proposed ISDG explicitly models intra-class dissimilarity to capture intra-class diversity. In ISSG, neighboring samples with higher spectral-spatial similarity are assigned larger weight coefficients, thereby emphasizing intra-class cohesion. In contrast, in ISDG, neighboring samples with greater spectral-spatial dissimilarity are assigned larger weight coefficients. Notably, in ISDG, the spectral-spatial variability among samples, the larger corresponding weight coefficients. This ensures that greater attention is paid to intra-class differences, allowing the model to capture fine-grained variations within each class. By jointly modeling similarity and dissimilarity, the proposed approach yields more comprehensive feature representations and ultimately improves hyperspectral classification performance.

The resulting graph structures can be denoted as $\boldsymbol{G}^L_{ISSG} = (\boldsymbol{V}^L, \boldsymbol{A}^L_{ISSG})$ and $\boldsymbol{G}^L_{ISDG} = (\boldsymbol{V}^L, \boldsymbol{A}^L_{ISDG})$, where $\boldsymbol{V}^L$ consists of a set of nodes representing individual samples, $\boldsymbol{A}^L_{ISSG}$ and $\boldsymbol{A}^L_{ISDG}$ denote the adjacency matrices of ISSG and ISDG, respectively. In ISSG, edges emphasize relationships among highly similar intra-class samples, with weights typically defined using Euclidean distance based on spectral-spatial information. This graph highlights the consistency and cohesiveness of samples within a class. In contrast, ISDG emphasizes the diversity and dissimilarity among intra-class samples, with edge weights defined using a novel dissimilarity distance measure. For the test sample set $\boldsymbol{X}^U$, a dedicated graph structure is constructed using a comprehensive integration strategy. First, the test sample set is augmented by incorporating samples from a spatial window centered on each test sample. Then, the augmented testing sample set is combined with the labeled training sample set $\boldsymbol{X}^L$ to form the complete sample set under consideration, denoted by $\boldsymbol{X}^A$, with duplicate samples removed. Since the semantic information of testing samples is unavailable, the graph structure $\boldsymbol{G}^A$ for this sample set prioritizes neighboring samples with high spectral-spatial similarity. The proposed separated graph construction scheme fully exploits the available information to capture both commonalities and differences among samples, thereby laying a solid foundation for subsequent feature extraction and land-cover classification.

### B. Dual-Branch GCN Model With Contrastive Learning

Although spectral variability and spatial differences exist within the same scene, land-cover samples belonging to the same category often exhibit consistency in high-order feature representations. On this basis, we propose a dual-branch GCN framework composed of a main branch and an auxiliary branch. The main branch is built upon the ISSG structure and serves as the primary network for feature extraction and classification. Meanwhile, the auxiliary branch leverages the ISDG structure to assist the main branch in extracting more robust landcover features and improving the model's generalization performance. During the testing phase, only the main branch is utilized for label prediction. The propagation rule in the main branch can be formulated as follows:

$$\begin{aligned}\tilde{\boldsymbol{H}}^{L(l)}_{ISSG} &= \mathrm{ReLU}\left(\hat{\boldsymbol{H}}^{L(l)}_{ISSG}\boldsymbol{W}^{L(l)}_{ISSG}+\boldsymbol{b}^{L(l)}_{ISSG}\right)\\ &= \mathrm{ReLU}\left(\left(\widetilde{\boldsymbol{D}}^L_{ISSG}\right)^{-1/2}\widetilde{\boldsymbol{A}}^L_{ISSG}\left(\widetilde{\boldsymbol{D}}^L_{ISSG}\right)^{-1/2}\right.\\ &\qquad \left.\tilde{\boldsymbol{H}}^{L^{(l-1)}}_{ISSG}\boldsymbol{W}^{L(l)}_{ISSG}+\boldsymbol{b}^{L(l)}_{ISSG}\right)\end{aligned} \tag{2}$$

where $\tilde{\boldsymbol{H}}^{L(l)}_{ISSG}$ denotes the learned feature map of the $l$th graph convolution layer, with $\tilde{\boldsymbol{H}}^{L(0)}_{ISSG} = \boldsymbol{X}^L$ and $\boldsymbol{W}^{L(l)}_{ISSG}$ representing the weight matrix of the $l$th layer. $\widetilde{\boldsymbol{A}}^L_{ISSG} = \boldsymbol{A}^L_{ISSG} + \boldsymbol{I}$ is the renormalization adjacency matrix, where $\boldsymbol{A}^L_{ISSG}$ is the original adjacency matrix and $\boldsymbol{I}$ is the identity matrix. $\widetilde{\boldsymbol{D}}^L_{ISSG}$ is the corresponding degree matrix, where each diagonal element denotes the row sum of the renormalized adjacency matrix $\widetilde{\boldsymbol{A}}^L_{ISSG}$. This renormalization operation improves the stability of graph convolution. After two graph convolutional layers, the learned spatial-spectral features are fed into a softmax classifier for label prediction. The preliminary classification result of the main branch is expressed as follows:

$$\hat{\boldsymbol{Y}}_{ISSG} = \mathrm{softmax}\left(f_{GCN2}\left(f_{GCN1}\left(\tilde{\boldsymbol{H}}^{L(0)}_{ISSG}\right)\right)\right). \tag{3}$$

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Similarly, the auxiliary branch adopts the same network architecture as the main branch, but operates on the ISDG. Importantly, the parameters of the two branches are trained independently and are not shared.

As illustrated in Fig. 2, the proposed SICM framework performs training and testing separately, which differs from conventional graph-based methods. During training, the total loss consists of the classification losses of the two branches, the cross-branch contrastive loss, and the regularization term. These components are jointly optimized to train the networks [58], [59]. The auxiliary branch improves the main branch by enhancing feature consistency, thereby promoting better classification performance. The overall loss function is defined as follows:

$$\mathcal{L} = \mathcal{L}_{CE} + \alpha \mathcal{L}_{CL} + \beta \mathcal{L}_{reg} \tag{4}$$

where $\mathcal{L}_{CE}$ denotes the mean cross-entropy classification loss of the two branches, $\mathcal{L}_{CL}$ denotes the cross-branch contrastive loss, and $\mathcal{L}_{reg}$ denotes the $L_2$-norm regularization term. The hyperparameter $\beta$ controls the strength of regularization, whereas the contrastive loss weight $\alpha$ balances the contribution of contrastive learning. The mean classification loss $\mathcal{L}_{CE}$ is computed as follows:

$$\mathcal{L}_{CE} = -\frac{1}{N^L} \sum_{i=1}^{N^L} \sum_{c=1}^{C} y_{i,c}^L \left( \log \hat{y}_{i,ISSG} + \log \hat{y}_{i,ISDG} \right) \tag{5}$$

where $y_{i,c}^L$ is the one-hot encoded ground truth label indicating whether sample $\boldsymbol{x}_i^L$ belongs to class $c$, $\hat{y}_{i,ISSG}$ and $\hat{y}_{i,ISDG}$ denote the predicted class probabilities produced by the main and auxiliary branches, respectively. $N^L$ is the number of labeled samples, and $C$ is the number of classes. By combining predictions of both branches, the classification loss can exploit complementary information captured by ISSG and ISDG. The cross-branch contrastive loss $\mathcal{L}_{CL}$ encourages feature similarity for samples from the same class while promoting dissimilarity for samples from different classes. Mathematically, it is defined as follows:

$$\mathcal{L}_{CL} = -\frac{1}{N^L} \sum_{i=1}^{N^L} \sum_{j=1}^{N^L} \times \left[ \Gamma_{ij} \log \delta \left( \frac{S_{ij}}{\tau} \right) + \left(1 - \Gamma_{ij}\right) \log \delta \left( -\frac{S_{ij}}{\tau} \right) \right] \tag{6}$$

where $\delta(x) = 1/(1 + e^{-x})$ denotes the sigmoid function, $\tau$ is the temperature parameter controlling the sharpness of the similarity distribution, empirically set to 0.5, and $\Gamma_{ij}$ is an indicator function that equals 1 when samples $\boldsymbol{x}_i^L$ and $\boldsymbol{x}_j^L$ belong to the same class [59], [60]. $S_{ij}$ represents the cosine similarity between the feature representations extracted from the two branches, i.e., between $\boldsymbol{F}_{i,ISSG}^L$ and $\boldsymbol{F}_{j,ISDG}^L$. This formulation encourages same-class samples to be close in the latent space while pushing different-class samples farther apart.

During training, a dynamic weighting mechanism is further introduced to adaptively balance the contributions of different losses to achieve different learning objectives at different stages [59], [61]. Specifically, the contrastive loss weight $\alpha$ is designed using a dual-component strategy that combines curriculum scheduling and loss-magnitude balancing. The base weight $\alpha_{base}$ follows a sinusoidal schedule:

$$\alpha_{base}(t) = \alpha_0 + \eta \sin\left(\frac{\pi t}{T}\right) \tag{7}$$

where $t$ denotes the current training epoch, $T$ is the total number of training epochs, $\alpha_0$ is the baseline weight, and $\eta$ controls the modulation amplitude [62]. This strategy prevents the model from overemphasizing contrastive regularization too early, gradually increasing its contribution as feature representations become more stable, and then shifting focus back toward classification in the later stages. Moreover, to alleviate scale imbalance between $\mathcal{L}_{CE}$ and $\mathcal{L}_{CL}$ during optimization, a loss-adaptive balancing factor $\zeta(t)$ is introduced, based on the exponential moving averages (EMAs) of the loss magnitudes in logarithmic space, as follows:

$$\zeta(t) = \text{clip}\left(\exp\left(\text{EMA}_t\left(\mathcal{L}_{CE}\right) \text{-EMA}_t\left(\mathcal{L}_{CL}\right)\right), \zeta_{\min}, \zeta_{\max}\right) \tag{8}$$

where $\text{EMA}_t(\mathcal{L}) = \rho \cdot \text{EMA}_{t-1}(\mathcal{L}) + (1 - \rho) \cdot \log(\mathcal{L} + \varepsilon)$ denotes the EMA at epoch $t$ with momentum coefficient $\rho$ controlling temporal smoothing and a small numerical stabilizer $\varepsilon$ to avoid singularities [63]. $[\zeta_{\min}, \zeta_{\max}]$ denotes the clipping bounds to prevent extreme adjustments. When $\mathcal{L}_{CL}$ is much larger than $\mathcal{L}_{CE}$, the log difference becomes negative, reducing $\zeta(t)$ toward its lower bound $\zeta_{\min}$; when the losses are comparable, $\zeta(t)$ increases toward $\zeta_{\max}$, thereby strengthening the contribution of contrastive learning. The final adaptive contrastive weight $\alpha$ is given by the following equation:

$$\alpha(t) = \alpha_{base}(t) \cdot \zeta(t). \tag{9}$$

This dual-component mechanism creates a curriculum-driven optimization process that naturally adapts to training dynamics. In the early training stage, the representations learned by the two branches are poorly aligned because of random initialization and different graph topologies. As a result, $\mathcal{L}_{CL}$ is relatively large compared with $\mathcal{L}_{CE}$, which drives the balancing factor $\zeta(t)$ toward its lower bound $\zeta_{\min}$. Even though $\alpha_{base}(t)$ gradually increases, the overall weight $\alpha(t)$ remains relatively small, allowing the model to first learn discriminative features under supervised signals. This mechanism prevents premature feature alignment, which could otherwise damage decision boundaries when class prototypes are still unstable. As training proceeds, intra-branch representations become more mature and $\mathcal{L}_{CL}$ decreases, indicating improved representation quality. Consequently, $\zeta(t)$ increases toward $\zeta_{\max}$. Meanwhile, $\alpha_{base}(t)$ reaches its sinusoidal peak, and together they amplify $\alpha(t)$, thereby strengthening cross-branch contrastive alignment. This enhanced alignment enforces consistency between the ISDG-derived features of the auxiliary branch and the ISSG-derived features of the main branch. In the later training stage, both $\mathcal{L}_{CE}$ and $\mathcal{L}_{CL}$ converge to low and stable values, indicating that the feature spaces are well aligned and the decision boundaries are sufficiently refined. At this point, $\zeta(t)$ stabilizes within a moderate range, while the decreasing $\alpha_{base}(t)$ gradually reduces $\alpha(t)$, allowing the optimization process to focus more on classification loss and fine-grained boundary refinement while preserving the learned inter-branch consistency.

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TABLE I
MAIN INFORMATION OF EACH DATASET

| Dataset | Indian Pines | Pavia University | Houston University | Trento |
|---|---|---|---|---|
| Collection area | Indiana | Pavia | Houston | Trento |
| Sensor type | AVIRIS | ROSIS | CASI 1500 | AISA Eagle |
| Image size | 145×145 | 610×340 | 349×1905 | 166×600 |
| NO. bands | 200 | 103 | 144 | 63 |
| Spectral range(μm) | 0.4-2.5 | 0.43-0.86 | 0.38-1.05 | 0.42-0.99 |
| Spatial resolution(m) | 20 | 1.3 | 2.5 | 1 |
| NO. classes | 16 | 9 | 15 | 6 |
| NO. standard training samples | 695 | 3921 | 2832 | 819 |
| NO. standard test samples | 9671 | 40002 | 12197 | 29395 |

It should be emphasized that the auxiliary branch is not directly used during testing. Instead, its role is to provide additional constraints and complementary information during training to improve the robustness of the main branch. By doing so, the main branch can better capture both intra-class similarity and intra-class variability, enabling it to model complex feature relationships more effectively. During testing, only the main branch is used to predict the labels of unknown samples. The output of the main branch on the concerned sample set $\boldsymbol{X}^A$ yields the initial classification result $\boldsymbol{Y}^A$.

### C. Local Context-Based Class Label Refinement

Considering the dense spatial distribution of ground objects belonging to the same category, we further employ a statistical analysis of class labels within a local window around each test sample [54], [64]. A majority voting strategy is used to determine the dominant semantic category within the local homogeneous region, and this information is then used to adaptively refine the label of the central sample, thereby producing the final classification result. For an unknown sample $\boldsymbol{x}_i^U \in \boldsymbol{X}^U$, let its initial predicted label obtained from $\boldsymbol{Y}^A$ be denoted by $y_i^U$. Let $\mathrm{N}(\boldsymbol{x}_i^U)$ denote the set of samples contained in a local window centered on $\boldsymbol{x}_i^U$ with spatial size $d_{win} \times d_{win}$. Thus, $d_{win}^2$ is the total number of samples in $\mathrm{N}(\boldsymbol{x}_i^U)$ and $\boldsymbol{Y}^W$ denotes the corresponding set of class labels within this window.

For samples in the local window that belong to the training set, their ground truth labels are used directly; otherwise, their initial predicted labels from the previous stage are adopted. The category of the central sample $\boldsymbol{x}_i^U$ within the local window is then refined using the following equation:

$$y_i^{U'} = \begin{cases} \arg\max_c \sum_{k=1}^{d_{win}^2} \Gamma_{\boldsymbol{Y}_k^W = c}, & \text{if } \max \sum_{k=1}^{d_{win}^2} \Gamma_{\boldsymbol{Y}_k^W = c} > \left(d_{win}^2/2\right) \\ y_i^U, & \text{else} \end{cases} \tag{10}$$

where $\Gamma_{\boldsymbol{Y}_i^W = c}$ is an indicator function that equals 1 if the category label $\boldsymbol{Y}_k^W$ of the $k$th sample in the $\mathrm{N}(\boldsymbol{x}_i^U)$ in the window belongs to class $c$, and 0 otherwise. Building upon the previous stage, which exploits long-range correlations among land-cover samples and captures both variability and consistency among known labeled samples, this local spatial-context refinement strategy further improves classification reliability and produces the final land-cover map.

**Algorithm 1** Pipeline of the Proposed Method

**Input:** The labeled sample set $\boldsymbol{X}^L \in \mathbb{R}^{N^L \times d_{SS}}$ with class label $\boldsymbol{Y}^L$, the test sample set $\boldsymbol{X}^U \in \mathbb{R}^{N^U \times d_{SS}}$, both obtained from the re-expressed hyperspectral data in the spectral-spatial domain.

1: For each labeled sample in $\boldsymbol{X}^L$, identify its intra-class $k$-nearest neighbors and intra-class $k$-farthest neighbors.
2: Construct two graph structures for $\boldsymbol{X}^L$, namely ISSG and ISDG, by redefining the adjacency weights between according to Eq. 1.
3: Construct the concerned sample set $\boldsymbol{X}^A$ by combining the samples within a spatial window centered on each test sample with $\boldsymbol{X}^L$ and $\boldsymbol{X}^U$, removing duplicate samples, and then build the corresponding graph structure $\boldsymbol{G}^A$.
4: Build the dual-branch GCN model according to Eqs. 2-9, where the ISSG-based branch acts as the primary classifier.
5: Train the model using a predefined learning rate and number of epochs, and use the main branch with $\boldsymbol{X}^A$ and $\boldsymbol{G}^A$ to obtain the initial label predictions.
6: Refine the class label of each test sample $\boldsymbol{x}_i^U$ by Eq. 10.

**Output:** The refined label result $\boldsymbol{Y}^{U'}$ for $\boldsymbol{X}^U$.

Subsequently, a spatial-context-based refinement strategy is applied to exploit the semantic consistency of land-cover samples within local spatial neighborhoods. By analyzing the dominant class in each local window, the labels of test samples can be adaptively corrected. Overall, the proposed method progressively integrates various information through a hierarchical information fusion strategy and captures intrinsic land-cover sample correlations from different perspectives, thereby boosting land-cover classification performance. For clarity, the complete pipeline of the proposed method is summarized in Algorithm 1.

## IV. EXPERIMENTS AND ANALYSIS

### A. Datasets and Experimental Settings

To validate the generalization capability of the proposed method, four publicly available hyperspectral datasets that are widely used in the remote sensing community are employed in this article. Each dataset provides standard training and testing sets, enabling fair and reproducible performance evaluation. These datasets were collected by different sensors over different geographic regions, and their main characteristics

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are summarized in the corresponding table. The landcover categories, color codes, and spatial distributions of different ground objects are presented in Table I.

It is worth noting that the ground-object classes in the standard training sets are distributed in contiguous regions, which is more consistent with real-world annotation scenarios. In practice, labeled samples obtained through field surveys are usually spatially clustered rather than randomly scattered across the scene. By contrast, randomly sampling training pixels from the ground-truth map often produces spatially dispersed labeled samples, which is less realistic and may introduce additional variability into performance evaluation across different runs. Therefore, we adopt the predefined standard training-test splits provided with these benchmark datasets to ensure objective, reproducible, and consistent comparisons. The four representative datasets, collected from different areas and sensors, cover a wide range of landcover categories. The false-color composites and ground-truth maps of the four datasets are shown in Figs. 4–7.

1) Indian Pines (IPs): Acquired by NASA's AVIRIS sensor over agricultural fields in northwestern Indiana, this dataset consists of 145 × 145 pixels with 200 spectral bands covering 400–2500 nm at a spatial resolution of 20 m. It contains 16 distinct classes with 10,249 labeled samples. Since many classes correspond to spectrally similar vegetation types, this dataset poses substantial challenges for fine-grained crop classification.
2) Pavia University (PU): Captured by the ROSIS sensor over the urban area of Pavia in Northern Italy, this dataset contains 610×340 pixels with 103 spectral bands (430–860 nm) at 1.3 m spatial resolution. It features nine classes of predominantly urban materials, with particularly challenging spectral similarities between various man-made structures (e.g., gravel/bricks, asphalt roads/tar roofs) that often lead to misclassification in traditional approaches.
3) Houston University (HU): Collected using the CASI-1500 spectrometer over the HU campus and surrounding urban areas, this dataset consists of 349 × 1905 pixels with 144 spectral bands (380–1050 nm) at 2.5 m spatial resolution. It encompasses 15 landcover classes, with notable classification challenges arising from significant spectral variations within identical class types across the image caused by cloud shadows.
4) Trento Dataset: Acquired by the AISA Eagle sensor over rural southern Trento, Italy, this dataset comprises 160 × 660 pixels with 63 spectral bands (420–990 nm) at a spatial resolution of 1 m. It contains six annotated landcover classes. Its main difficulty lies in the presence of spectrally similar vegetation categories, such as apple trees, vineyards, and other trees, as well as built structures such as buildings and roads located in close spatial proximity.

Our experimental evaluation was conducted using the standard training and testing partitions to ensure fair comparison among different methods. All reported results are averaged over five independent runs to reduce the influence of random initialization. Performance was evaluated through both visual inspection and quantitative analysis using four widely adopted metrics: 1) class-specific accuracy (CA); 2) overall accuracy (OA); 3) average accuracy (AA); and 4) kappa coefficient ($\kappa$). Together, these metrics provide a comprehensive assessment of classification performance for both dominant and minority classes. For the EMAP-based texture feature extraction module in the proposed method, four attribute parameters were selected according to previous studies: area, diagonal length of the bounding box, standard deviation, and moment of inertia. Their value ranges were empirically set following [53], [54]. For clarity in the result tables, the best performance values are highlighted in **bold**, whereas the second-best values are indicated in *italics*.

### B. Comparative Experiments and Discussion

To thoroughly evaluate the effectiveness of the proposed method for HIC, we conducted comparative experiments against several state-of-the-art approaches, including GSPSA [34], $S^2$FTNet [35], $S^2$Mambat [36], $S^2$VNet [37], DSFormer [38], miniGCN [41], GraphGST [48], NESSGGCN [50], and WSGraph [51], which are representative methods related to GCNs, Transformers, and contrastive learning. All comparative methods were implemented using their optimal parameter configurations, as specified in the corresponding literature and open-source implementations, to ensure a fair comparison. In the graph construction process of the proposed SICM method, the number of nearest-neighbor samples was empirically set to 5 [41], [53]. In the proposed dual-branch architecture, the two branches share the same structural configuration but do not share parameters. In each branch, the graph convolutional layer outputs 96-D feature representations, which are subsequently transformed to 64-D embeddings through fully connected layers. The auxiliary branch is introduced to enhance the robustness of the main branch during training. The dual-branch GCN model is trained for 300 epochs with a learning rate of 0.001. For the total loss function defined in (4), the regularization coefficient $\beta$ is fixed at 0.001 to prevent overfitting. For the adaptive contrastive loss weighting mechanism described in (7)–(9), the key parameters are set as $\alpha_0 = 0.2, \eta = 0.4$, EMA momentum coefficient $\rho = 0.9$, numerical stabilizer $\varepsilon = 1e-8$, and $[\zeta_{\min}, \zeta_{\max}] = [0.2, 2.0]$. Through this adaptive weighting mechanism, the contrastive coefficient $\alpha(t)$ dynamically balances the contributions of the supervised classification loss and the contrastive alignment loss according to their relative magnitudes during training, thereby ensuring effective collaboration between the dual-branch architecture and the consistency regularization objective throughout the optimization process. Importantly, during inference on unknown test samples, only the main branch is used for final class-label prediction, thereby maintaining computational efficiency while retaining the benefits of the dual-branch training paradigm. The overall computational complexity of the proposed method is mainly determined by the cost of graph construction and the number of training epochs.

*1) Quantitative Comparison and Analysis of IP:* Table II presents the quantitative performance metrics of all evaluated methods on the IP dataset. The results demonstrate that the

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TABLE II
CA, OA, AA (%), AND $k$ ($\times 100$) OBTAINED BY DIFFERENT METHODS ON THE IP DATASET

| Class code | No. Train/Test | GSPSA | S$^2$FTNet | S$^2$Mamba | S$^2$VNet | DSFormer | miniGCN | GraphGST | NEGCN | WSGraph | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 50/1384 | 97.6744 | 86.3439 | 97.2367 | 92.6346 | 87.3861 | 84.8705 | 98.2031 | **98.9891** | 91.2698 | *98.6272* |
| 2 | 50/784 | 90.3915 | 95.6633 | *97.0297* | 85.4237 | 83.9168 | 49.6923 | 92.7500 | 94.2238 | 88.0597 | **99.2347** |
| 3 | 50/184 | 88.0383 | *99.4565* | 94.8454 | 82.8829 | 81.2500 | 62.7737 | 90.1478 | 97.8723 | 91.0891 | **100** |
| 4 | 50/447 | *99.2806* | 93.5123 | **100** | 98.6577 | 94.5770 | 88.4615 | 92.4612 | **100** | 97.5169 | 98.6577 |
| 5 | 50/697 | 94.3166 | **100** | 99.1465 | 99.7131 | 95.3058 | 94.4759 | 99.1367 | *99.8567* | 99.1241 | **100** |
| 6 | 50/439 | *99.7674* | 99.3166 | **100** | **100** | **100** | 98.6425 | 99.5444 | **100** | **100** | **100** |
| 7 | 50/918 | 74.4072 | 88.6710 | *96.5885* | 91.8346 | 88.0829 | 52.8846 | 88.0478 | 95.7128 | 93.7136 | **99.4553** |
| 8 | 50/2418 | 97.8072 | 83.8296 | *98.2809* | 97.1520 | 97.5195 | 86.7485 | 96.6867 | 97.1487 | 98.1780 | **99.7932** |
| 9 | 50/564 | 84.2193 | 90.0709 | 97.6407 | 96.9091 | 96.1315 | 59.5376 | 91.3706 | *99.0909* | 93.9850 | **99.1135** |
| 10 | 50/162 | 85.7143 | 98.7654 | **100** | 98.1818 | 90.0000 | 94.1520 | 97.5610 | *99.3827* | **100** | *99.3827* |
| 11 | 50/1244 | 99.5070 | *99.6785* | 99.1903 | 99.2689 | 96.6035 | 98.3943 | 99.3475 | 99.5165 | **99.8392** | **99.8392** |
| 12 | 50/330 | 87.6404 | **100** | 93.4844 | 90.3581 | 83.3333 | 70.0495 | 89.6936 | *95.3757* | 93.7143 | **100** |
| 13 | 50/45 | 70.3125 | **100** | 75.0000 | 63.3803 | 77.5862 | 84.9057 | 78.9474 | *86.5385* | 83.3333 | **100** |
| 14 | 15/39 | **100** | 87.1795 | 88.6364 | 92.8571 | 90.6977 | 87.5000 | *97.5000* | 95.0000 | 74.5098 | 97.4359 |
| 15 | 15/11 | 34.3750 | **100** | 64.7059 | *73.3333* | 57.8947 | 55.0000 | 47.8261 | 61.1111 | **100** | **100** |
| 16 | 15/5 | **100** | **100** | *83.3333* | 38.4615 | 83.3333 | 41.6667 | 33.3333 | **100** | 20.8333 | **100** |
| OA (%) | — | 92.2345 | 91.5417 | 97.6941 | 94.5921 | 92.0794 | 75.7005 | 95.0884 | *97.7252* | 95.3056 | **99.4830** |
| AA (%) | — | 87.7157 | *95.1555* | 92.8199 | 87.5655 | 87.7262 | 75.6097 | 87.0348 | 94.9887 | 89.0729 | **99.4712** |
| $K \times 100$ | — | 91.1395 | 90.3432 | 97.3540 | 93.8196 | 90.9681 | 87.4162 | 94.3749 | *97.3862* | 94.6149 | **99.1297** |

TABLE III
CA, OA, AA (%), AND $k$ ($\times 100$) OBTAINED BY DIFFERENT METHODS ON THE PU DATASET

| Class code | No. Train/Test | GSPSA | S$^2$FTNet | S$^2$Mamba | S$^2$VNet | DSFormer | miniGCN | GraphGST | NEGCN | WSGraph | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 548/6304 | 99.1678 | 86.8813 | *99.4801* | 99.1956 | 98.8614 | 96.4571 | 72.2811 | 99.1446 | 97.9287 | **99.9524** |
| 2 | 540/18146 | 97.1895 | 70.9302 | *99.9830* | 97.8526 | 99.9727 | 96.8297 | 96.9667 | 94.5237 | 98.9753 | **100** |
| 3 | 392/1815 | 91.7148 | 76.7493 | *98.2955* | 95.3503 | 91.6412 | 84.0708 | 91.0323 | 97.6245 | 91.0349 | **100** |
| 4 | 524/2912 | 63.4267 | 70.5701 | 95.1949 | *95.8913* | 89.4200 | 85.1619 | 84.5980 | 68.2724 | 79.6500 | **99.6566** |
| 5 | 265/1113 | 97.3730 | 98.8320 | 99.2806 | *99.5491* | 99.9093 | 99.3699 | 91.4407 | **100** | 97.2052 | **100** |
| 6 | 532/4572 | 59.3915 | *93.8320* | 91.6986 | 90.0406 | 58.0150 | 90.7811 | 53.1034 | 92.5618 | 57.9017 | **100** |
| 7 | 375/981 | 90.0000 | 97.3496 | 96.9368 | 98.5552 | *98.7780* | 69.9743 | 73.1446 | 93.1596 | 67.6364 | **100** |
| 8 | 514/3364 | 82.5730 | *94.8276* | 85.2467 | 88.9220 | 82.8439 | 72.2482 | 74.5636 | 74.7495 | 85.6503 | **100** |
| 9 | 231/795 | 99.1206 | 94.5912 | 94.0547 | 93.2704 | 98.5915 | 97.0207 | 96.9363 | *99.7475* | 86.2616 | **100** |
| OA (%) | — | 85.3057 | 80.2035 | *96.7952* | 96.0477 | 88.6531 | 91.1979 | 79.4335 | 90.3755 | 86.4282 | **99.9950** |
| AA (%) | — | 86.6619 | 87.1737 | *95.5745* | 95.4030 | 90.8926 | 87.9904 | 81.5630 | 91.0871 | 84.6938 | **99.9949** |
| $K \times 100$ | — | 92.2532 | 74.7479 | *95.6984* | 94.6707 | 85.2754 | 89.3529 | 73.7335 | 87.1033 | 82.4032 | **99.9949** |

proposed method consistently outperforms the comparative approaches across all evaluation metrics.

Specifically, the proposed method achieves an OA of 99.4830%, representing absolute improvements of 4.4%, 1.8%, and 4.2% over recently developed graph-based classification methods including GraphGST, NESSGGCN, and WSGraph, respectively. Despite the challenging nature of this dataset, which contains numerous spectrally similar crop and vegetation types, the proposed method demonstrates strong discrimination capability for ground objects with subtle spectral differences. Notably, for categories that are frequently misclassified by competing methods, such as Corn-min till (Class 3), Grass/Pasture-mowed (Class 7), and Oats (Class 9), the proposed method still achieves the highest CA values.

*2) Quantitative Comparison and Analysis of PU:* As shown in Table III, the proposed method achieves superior performance across all key evaluation metrics on the PU dataset. This urban dataset presents particular challenges because it contains diverse anthropogenic materials, including asphalt, bricks, concrete, and metal sheeting, with subtle spectral differences. Most classification approaches frequently suffer from confusion between similar classes, such as Gravel (Class 3) and Bricks (Class 8), making high-precision discrimination particularly difficult. The proposed method effectively overcomes these limitations and achieves the highest CA across all categories. Compared with state-of-the-art classification approaches such as S$^2$Mambat, S$^2$VNet, and DSFormer, the method yields substantial improvements in OA of 3.2%, 3.9%, and 11.3%, respectively, further demonstrating its effectiveness for artificial-material classification.

*3) Quantitative Comparison and Analysis of Hu:* Table IV reports the classification accuracy of each method on the HU dataset. This dataset presents unique challenges due to cloud-cover effects, which introduce significant spectral variations within identical classes across building, road, and vegetation regions affected by cloud shadows. Despite these challenges, the proposed method achieves an exceptional OA of 98.8932%, together with consistently superior class-specific accuracies across multiple categories. Competing methods exhibit unbalanced performance across categories, with notable accuracy degradation in shadow-affected classes such as Railways (Class 11) and Parking trucks (Class 15). In contrast, the proposed method maintains high classification accuracy even for these challenging categories, demonstrating

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TABLE IV

CA, OA, AA (%), AND $k$ ($\times 100$) OBTAINED BY DIFFERENT METHODS ON THE HU DATASET

| Class code | No. Train/Test | GSPSA | S$^2$FTNet | S$^2$Mamba | S$^2$VNet | DSFormer | miniGCN | GraphGST | NEGCN | WSGraph | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 198/1053 | 99.6560 | 90.1235 | **100** | 99.8856 | 100 | *99.7719* | 95.9442 | 99.7605 | 86.8421 | 98.5755 |
| 2 | 190/1064 | 91.5392 | 93.1391 | **100** | 99.2325 | 98.3713 | 98.3678 | 85.2856 | 97.3913 | 78.5208 | *99.3421* |
| 3 | 192/505 | **100** | 55.2475 | 91.9414 | *98.8235* | 58.0000 | 42.2294 | 70.7174 | **100** | 82.1053 | **100** |
| 4 | 188/1056 | 97.3843 | 85.9848 | **100** | 99.7961 | 97.1774 | *99.8969* | 94.9735 | 77.4194 | 89.4309 | 98.7689 |
| 5 | 186/1056 | 98.9691 | 99.2424 | **100** | 98.3225 | *99.9054* | 98.5942 | 97.4954 | 97.5069 | 89.6324 | **100** |
| 6 | 182/143 | 10.6284 | 95.8042 | **99.2754** | 10.6222 | 78.8889 | 22.6667 | 33.3333 | 80.7143 | 63.7584 | *97.2028* |
| 7 | 196/1072 | 85.3095 | 77.1455 | 88.1801 | *91.8860* | 96.7230 | 63.7363 | 76.3876 | 53.2115 | 59.4648 | **99.7201** |
| 8 | 191/1053 | 75.2434 | 73.0294 | 63.5027 | 87.5616 | 80.1824 | 84.0491 | 74.4777 | **98.5314** | 58.6881 | *96.0114* |
| 9 | 193/1059 | 88.6517 | 83.6638 | 88.8995 | 74.8259 | *89.9901* | 81.5134 | 84.4350 | 66.0530 | 54.0366 | **98.9613** |
| 10 | 191/1036 | 80.7736 | 51.7375 | 84.5398 | 70.8564 | *96.6129* | 70.9068 | 79.2424 | 90.4051 | 80.4598 | **97.7799** |
| 11 | 181/1054 | 96.2773 | 85.2941 | **99.3296** | 94.4041 | 68.8073 | 82.7869 | 83.0645 | 76.1951 | 48.6182 | *99.0512* |
| 12 | 192/1041 | 87.5235 | 87.8963 | 95.5408 | 89.6648 | *96.7262* | 84.2629 | 54.2565 | 83.1878 | 80.4324 | **100** |
| 13 | 184/285 | 44.5629 | 88.0702 | *95.1493* | 82.2526 | 94.9275 | 78.4375 | 95.0673 | 75.8294 | 48.9676 | **97.5439** |
| 14 | 181/247 | 93.2075 | 97.5709 | **100** | 92.7203 | **100** | *98.6175* | 95.3307 | **100** | 39.0205 | **100** |
| 15 | 187/473 | *98.3015* | 80.9725 | 91.9922 | 97.5155 | 93.6634 | 95.5457 | 93.0886 | 93.2939 | 74.3068 | **100** |
| OA (%) | — | 81.0527 | 82.1431 | *90.8748* | 82.7909 | 88.6365 | 79.5523 | 78.6751 | 81.2741 | 68.8612 | **98.8932** |
| AA (%) | — | 83.2019 | 82.9948 | *93.2234* | 85.8913 | 89.9984 | 80.0922 | 80.8733 | 85.9666 | 68.9523 | **98.8638** |
| $K \times 100$ | — | 83.7503 | 80.6123 | *90.0938* | 81.4619 | 87.6819 | 81.5372 | 76.8983 | 79.6528 | 66.2811 | **98.9447** |

TABLE V

CA, OA, AA (%), AND $k$ ($\times 100$) OBTAINED BY DIFFERENT METHODS ON THE TRENTO DATASET

| Class code | No. Train/Test | GSPSA | S$^2$FTNet | S$^2$Mamba | S$^2$VNet | DSFormer | miniGCN | GraphGST | NEGCN | WSGraph | Proposed |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | 129/3905 | 92.7845 | 86.4789 | *99.8404* | 99.2037 | 99.7341 | 94.8900 | 81.8932 | 73.7851 | **100** | **100** |
| 2 | 125/2778 | 83.4648 | 85.1692 | 75.6239 | 81.4826 | 90.5953 | 78.3532 | 78.0290 | 77.4608 | *92.2027* | **99.7840** |
| 3 | 105/374 | 51.4731 | 56.6845 | 71.4008 | 82.9099 | 73.1373 | 69.6121 | 74.0443 | 47.0440 | *86.1111* | **100** |
| 4 | 154/8969 | 99.3637 | 94.8824 | *99.9330* | 99.8081 | 99.8206 | 98.7832 | 97.3351 | 99.7880 | 99.7310 | **99.9889** |
| 5 | 184/10317 | 99.5551 | 50.4895 | 99.4790 | 99.6520 | 99.1530 | 99.7245 | 92.3245 | *99.9769* | 95.5101 | **100** |
| 6 | 122/3052 | 94.3820 | 83.7156 | 93.2656 | *96.0531* | 93.2437 | 87.4259 | 83.5832 | 92.0969 | 93.0194 | **99.8034** |
| OA (%) | — | 95.4890 | 75.6217 | 95.8768 | 97.0369 | *97.5540* | 94.9685 | 89.5458 | 90.6753 | 96.6185 | **99.9558** |
| AA (%) | — | 86.8372 | 76.2367 | 89.9238 | 93.1849 | 92.6140 | 88.1315 | 84.5349 | 81.6920 | *94.4290* | **99.9294** |
| $K \times 100$ | — | 93.9738 | 69.2441 | 94.4783 | 96.0337 | *96.7234* | 90.6139 | 86.0890 | 87.7347 | 95.4625 | **99.7402** |

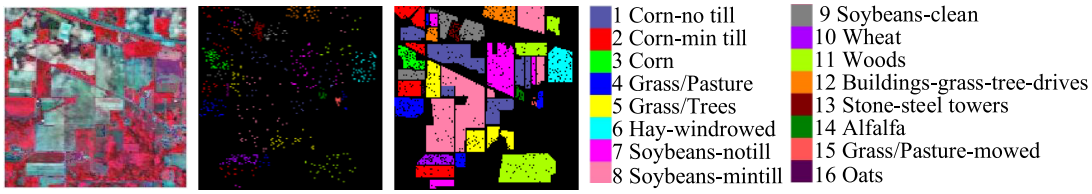



Fig. 4. Visualization of IP. (a) False-color image for HSI. (b) Standard training set in the ground truth map. (c) Standard test set in the ground truth map. (d) Color coding for each class.

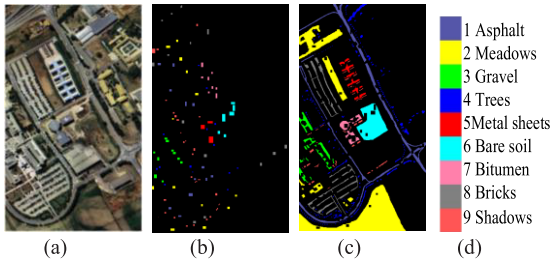



Fig. 5. Visualization of PU. (a) False-color image for HSI. (b) Standard training set in the ground truth map. (c) Standard test set in the ground truth map. (d) Color coding for each class.

that it can successfully extract robust features by effectively modeling intra-class sample associations and thereby effectively mitigate the adverse effects of shadow-induced spectral variations.

*4) Quantitative Comparison and Analysis of Trento:* Table V lists the quantitative results obtained by each classification method on the Trento dataset. The results show that the proposed method consistently achieves superior performance across the three commonly used classification metrics and exhibits balanced land-cover identification across all categories overall. For spectrally similar landcovers that are prone to misclassification, such as Apple trees (Class 1) and Vineyard (Class 5), the proposed method effectively differentiates them by leveraging spatial-spectral information to reveal the inherent correlations among samples.

*5) Visual Comparison of Classification Results:* To qualitatively evaluate the performance of different classification methods, we conducted visual experiments to complement the quantitative analysis. Figs. 8–11 present the classification maps generated by all comparative methods across the four datasets. The visual results clearly demonstrate that the

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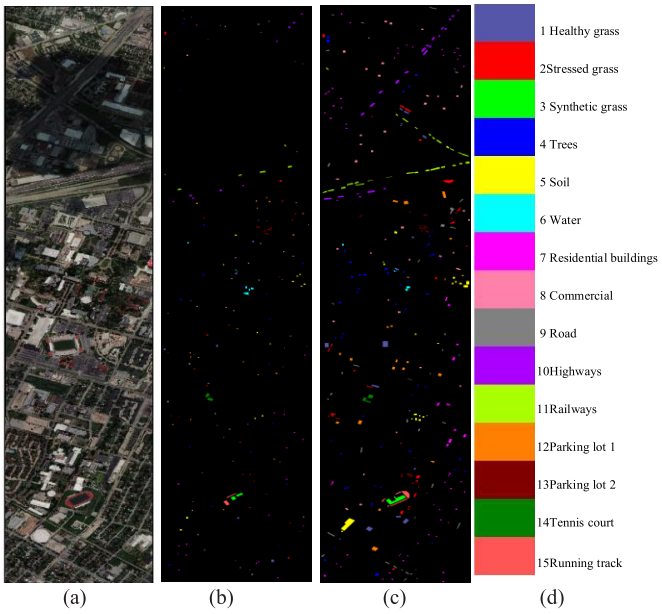



Fig. 6. Visualization of HU. (a) False-color image for HSI. (b) Standard training set in the ground truth map. (c) Standard test set in the ground truth map. (d) Color coding for each class.

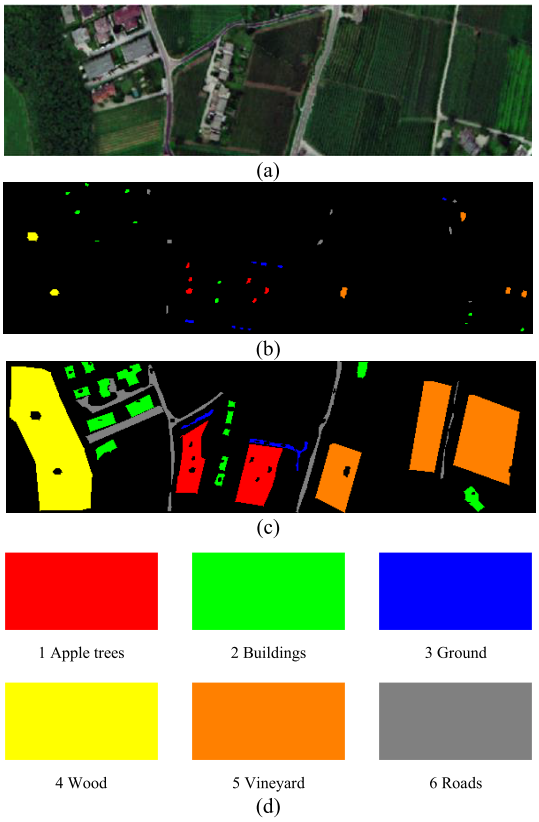



Fig. 7. Visualization of Trento. (a) False-color image for HSI. (b) Standard training set in the ground truth map. (c) Standard test set in the ground truth map. (d) Color coding for each class.

proposed method consistently produces classification maps with more precise boundaries, greater regional homogeneity, and substantially fewer misclassification artifacts than the alternative approaches. These visual improvements are consistent with the quantitative metrics reported in Tables II–V, providing further evidence of the superior performance of the proposed method. For example, in the IP dataset (Fig. 8), the proposed method successfully distinguishes different crop types with subtle spectral differences, whereas competing methods exhibit notable confusion among these closely related classes. Similar improvements can be observed in the other three datasets (Figs. 9–11), where the proposed method effectively discriminates among spectrally similar land-cover classes while preserving spatial integrity. In the HU dataset, regions affected by cloud occlusions pose significant challenges for HIC because shadows and occlusions distort the spectral signatures of ground objects. Nevertheless, the proposed method demonstrates strong discriminative capability for land-cover classes in these shadowed regions, where other classification methods show noticeable performance degradation.

It is worth noting that, in real-world remote sensing applications, annotated regions typically form continuous spatial blocks. In our experimental design, we utilized the standard training sets provided with each dataset as known semantic information, where labeled samples exhibit concentrated spatial distributions, with samples from each class typically originating from the same geographical region. Unlike randomly sampled training sets, which may inadvertently capture greater intra-class spectral variability, these standard training sets present more realistic yet challenging classification scenarios. Comparative methods generally show inconsistent performance on these standard datasets, with significant accuracy fluctuations across different land-cover types. The proposed method addresses this challenge by systematically leveraging intra-class spectral variability to guide robust feature extraction, together with a hierarchical spatial-spectral information fusion strategy. This overall framework significantly enhances both classification accuracy and cross-dataset generalization capability for HIC, as evidenced by the consistent performance improvements across all evaluated datasets. Overall, these visual assessments, together with the comprehensive quantitative analysis, provide compelling evidence for the effectiveness of the proposed method in addressing the challenges of HIC across diverse landscapes and environmental conditions.

### C. Ablation Study and Parameter Analysis

*1) Module Ablation:* In this section, we conducted a series of ablation experiments to evaluate the effectiveness of each component in the proposed SICM framework. Specifically, these experiments analyzed the contributions of the following key components: the re-expressed data representation, the network architecture, including the single GCN and the dual-branch GCN with different graph construction schemes (ISSG or ISDG), and the classification refinement mechanism based on local spatial context information. For a comprehensive and intuitive evaluation, we sequentially removed these components and denoted the resulting variants as SICM-v1, SICM-v2, SICM-v3, SICM-v4, SICM-v5, SICM-v6, and SICM-v7. The detailed configurations of these variants are illustrated in Fig. 12. All variants were evaluated under

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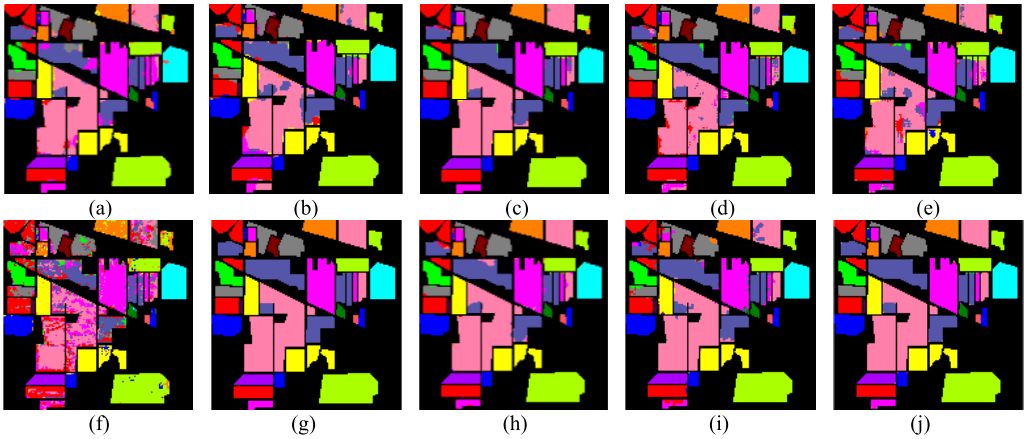



Fig. 8. Classification maps of each method on IP. (a) GSPSA. (b) $S^2$FTNet. (c) $S^2$Mambat. (d) $S^2$VNet. (e) DSFormer. (f) miniGCNs. (g) GraphGST. (h) NESSGGCN. (i) WSGraph. (j) Proposed.

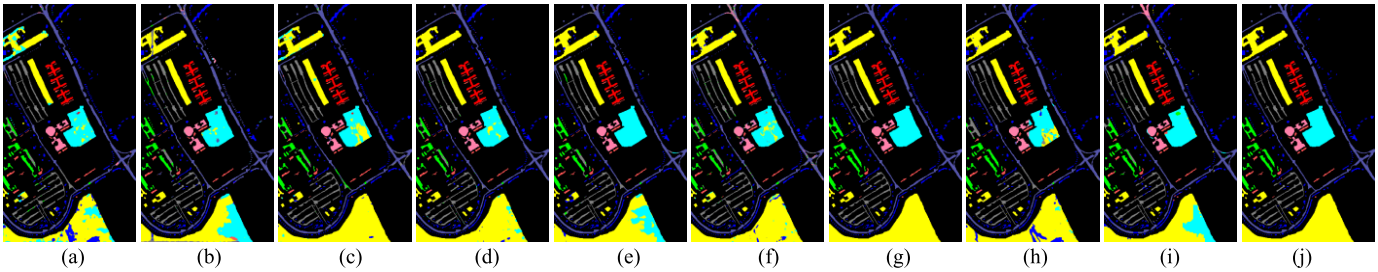



Fig. 9. Classification maps of each method on PU. (a) GSPSA. (b) $S^2$FTNet. (c) $S^2$Mambat. (d) $S^2$VNet. (e) DSFormer. (f) miniGCNs. (g) GraphGST. (h) NESSGGCN. (i) WSGraph. (j) Proposed.

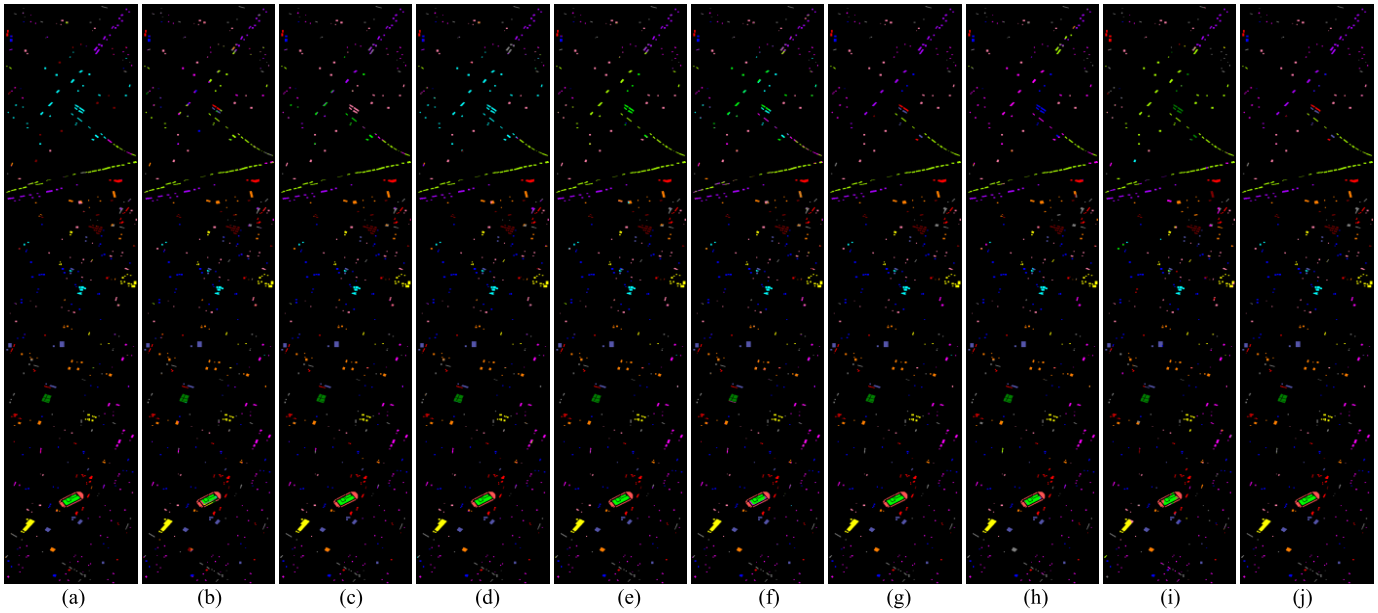



Fig. 10. Classification maps of each method on HU. (a) GSPSA. (b) $S^2$FTNet. (c) $S^2$Mambat. (d) $S^2$VNet. (e) DSFormer. (f) miniGCNs. (g) GraphGST. (h) NESSGGCN. (i) WSGraph. (j) Proposed.

the same parameter settings as the proposed method, and the corresponding results are reported in Table VI.

From the results, it can be observed that the single-branch method SICM-v2, which constructs the graph by prioritizing intra-class similarity among labeled samples, achieves substantially better classification performance than SICM-v3, which prioritizes intra-class discrepancy among labeled samples during graph construction. This result is consistent with the conventional GCN paradigm, in which mining sample similarity facilitates feature aggregation and alignment. It is also worth noting that the ISDG-based single-branch variant SICM-v3 yields considerably lower classification

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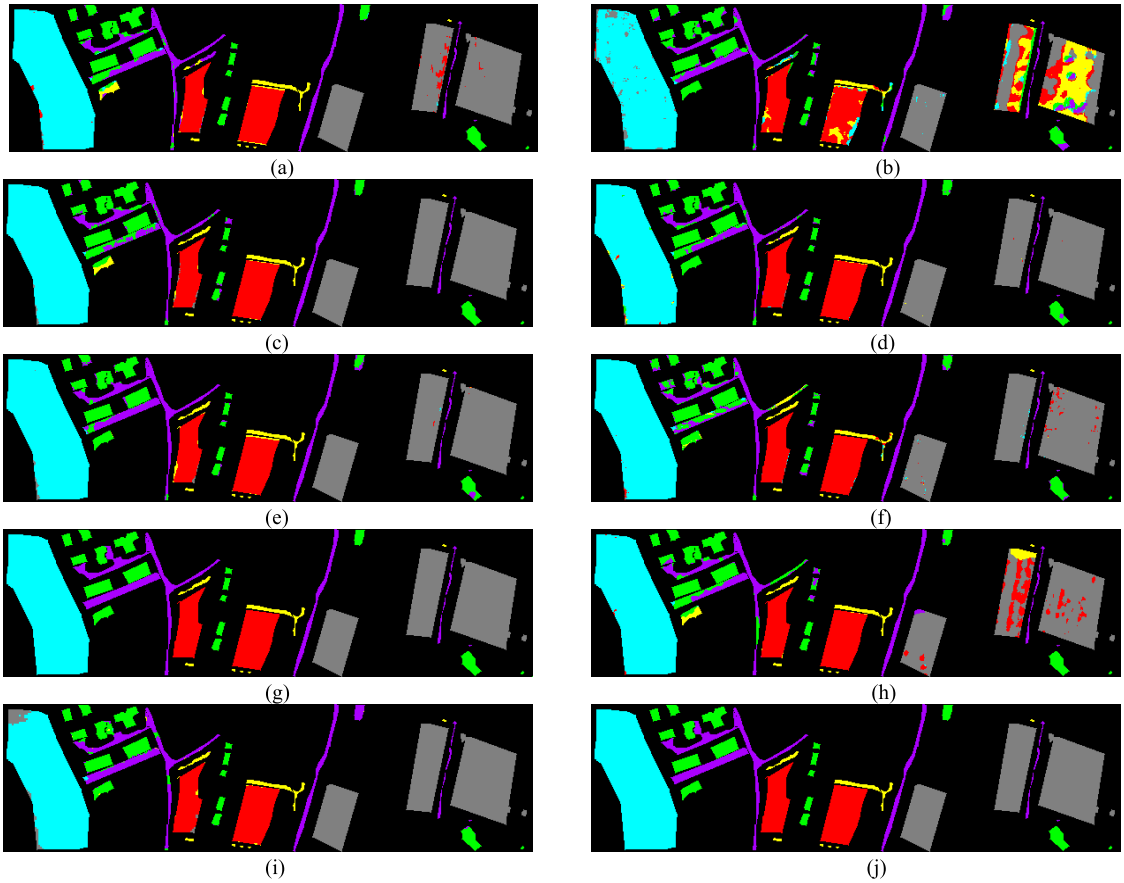



Fig. 11. Classification maps of each method on Trento. (a) GSPSA. (b) S$^2$FTNet. (c) S$^2$Mambat. (d) S$^2$VNet. (e) DSFormer. (f) miniGCNs. (g) GraphGST. (h) NESSGGCN. (i) WSGraph. (j) Proposed.

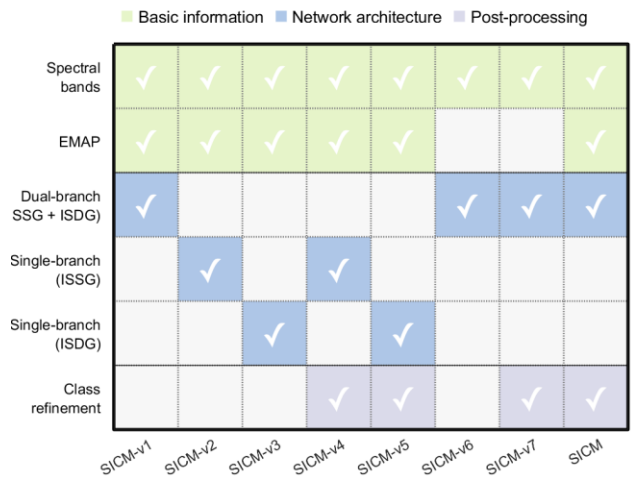



Fig. 12. Componentwise ablation analysis of SICM variants.

accuracy across all datasets. This outcome is expected because ISDG connects the most dissimilar intra-class samples, and its graph topology is not naturally suited for direct feature aggregation or classification. However, when incorporated into the dual-branch architecture as an auxiliary branch, ISDG contributes notable performance gains. The dual-branch method SICM-v1, which jointly incorporates ISSG and ISDG, consistently outperforms the single-branch SICM-v2 across all datasets, with OA improvements of 5.38% on IP, 0.36% on PU, 0.35% on HU, and 0.19% on Trento. These results indicate that the ISDG branch, through cross-branch contrastive learning, effectively transfers intra-class variability knowledge to the main branch without directly participating in classification, thereby enhancing the robustness of the learned feature representations. Furthermore, the performance gain brought by the ISDG branch varies across datasets, with more pronounced improvements observed on datasets exhibiting greater intra-class spectral variability, such as IP, which contains spectrally similar crop types, and HU, which is affected by cloud shadow-induced spectral variations. This finding suggests that the proposed dual-graph design is particularly beneficial in scenarios where intra-class discrepancy is more significant. Among the dual-branch GCN-based variants, SICM-v1 also performs noticeably better than SICM-v6, indicating that the collaborative use of spatial and spectral information can more effectively measure sample distances and capture the intrinsic relationships among samples, thereby improving classification performance. In addition, the classification refinement mechanism guided by local spatial-context information further improves classification accuracy, which further verifies the effectiveness of the proposed method.

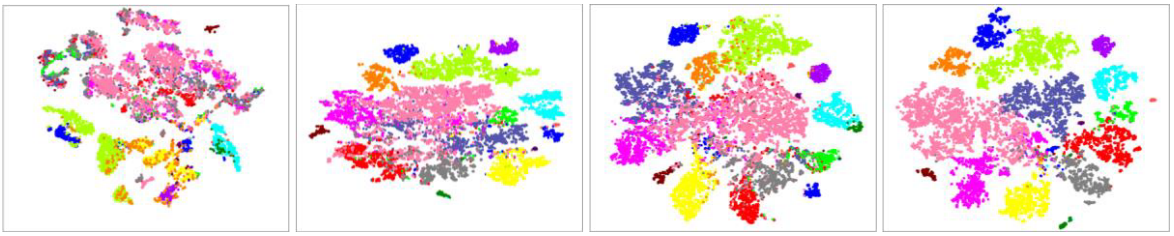
To further validate the feature extraction capability of the proposed dual-branch GCN model, we employed the t-stochastic neighbor embedding (t-SNE) technique to visualize the data distribution in the feature space for each dataset

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TABLE VI

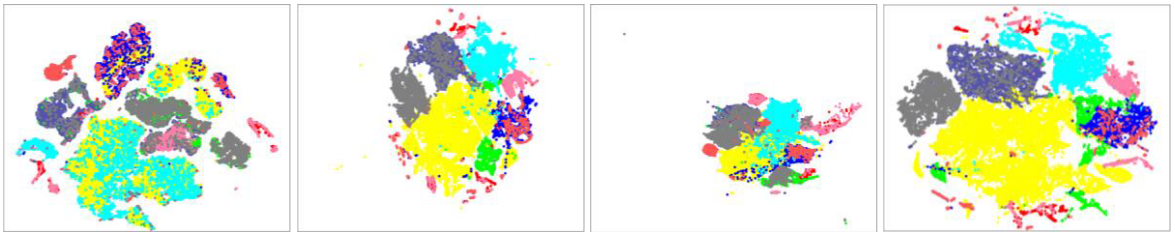
ABLATION RESULTS OF THE PROPOSED METHOD SCIM WITH A COMBINATION OF DIFFERENT COMPONENTS ON DIFFERENT DATASETS

| Metrics \ Method | | SICM-v1 | SICM-v2 | SICM-v3 | SICM-v4 | SICM-v5 | SICM-v6 | SICM-v7 | SICM |
|---|---|---|---|---|---|---|---|---|---|
| IP | OA (%) | 82.1321 | 76.7552 | 58.5048 | 99.4623 | 99.2969 | 58.9805 | 99.4313 | 99.4830 |
| | AA (%) | 88.4048 | 86.1518 | 70.5756 | 99.4627 | 99.3305 | 70.1496 | 99.4220 | 99.4712 |
| | $K \times 100$ | 79.6951 | 70.3413 | 50.5052 | 98.5219 | 93.1423 | 59.8351 | 98.0071 | 99.1297 |
| PU | OA (%) | 94.9053 | 94.5502 | 84.3233 | 99.9475 | 99.8650 | 79.0510 | 99.9575 | 99.9675 |
| | AA (%) | 96.3413 | 97.0652 | 85.8479 | 99.9219 | 99.7643 | 86.4131 | 99.9354 | 99.9566 |
| | $K \times 100$ | 94.7790 | 95.1025 | 78.5067 | 99.9392 | 99.6187 | 81.5361 | 99.9032 | 99.9574 |
| HU | OA (%) | 84.7176 | 84.3650 | 59.0637 | 98.8522 | 96.8353 | 62.6957 | 96.7779 | 98.8932 |
| | AA (%) | 87.0788 | 86.9098 | 64.6239 | 98.9766 | 97.1805 | 70.0445 | 97.2605 | 98.8638 |
| | $K \times 100$ | 86.7339 | 87.5509 | 63.4902 | 99.0380 | 95.9023 | 72.8521 | 95.6206 | 98.9447 |
| Trento | OA (%) | 96.3395 | 96.1456 | 86.2800 | 99.9524 | 99.9218 | 76.6423 | 99.9524 | 99.9558 |
| | AA (%) | 93.9202 | 93.2651 | 83.9736 | 99.9239 | 99.8781 | 79.8267 | 99.9278 | 99.9294 |
| | $K \times 100$ | 87.9645 | 88.8532 | 74.4900 | 99.7343 | 99.5097 | 72.0939 | 99.8535 | 99.7402 |



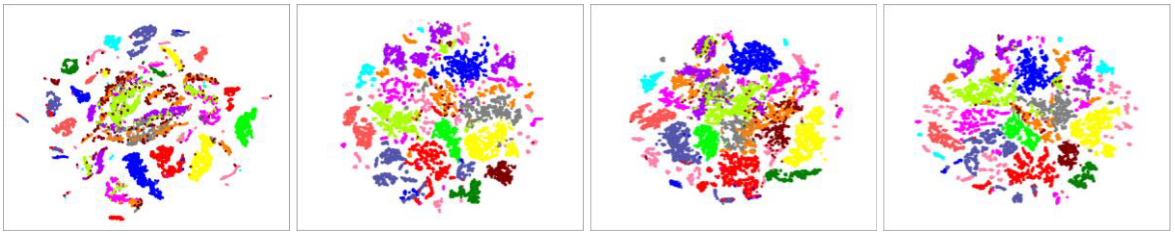
(a) (b) (c) (d)

Fig. 13. t-SNE feature visualization of class features obtained by different comparison methods on the IP dataset. (a) Class features output by PCA. (b) Class features output by SICM-v2 that considers the ISSG-based GCN and spectral-spatial information. (c) Class features output by SICM-v6 that considers the dual-branch GCN model and spectral information. (d) Class features output by SICM-v1 that considers the dual-branch GCN model and spectral-spatial information.



(a) (b) (c) (d)

Fig. 14. t-SNE feature visualization of class features obtained by different comparison methods on the PU dataset. (a) Class features output by PCA. (b) Class features output by SICM-v2 that considers the ISSG-based GCN and spectral-spatial information. (c) Class features output by SICM-v6 that considers the dual-branch GCN model and spectral information. (d) Class features output by SICM-v1 that considers the dual-branch GCN model and spectral-spatial information.

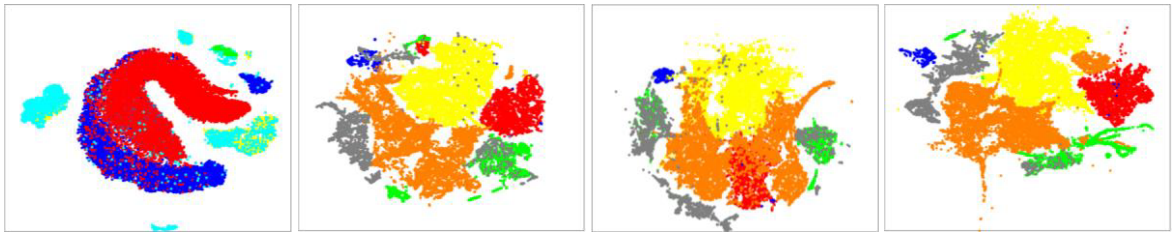


(a) (b) (c) (d)

Fig. 15. t-SNE feature visualization of class features obtained by different comparison methods on the HU dataset. (a) Class features output by PCA. (b) Class features output by SICM-v2 that considers the ISSG-based GCN and spectral-spatial information. (c) Class features output by SICM-v6 that considers the dual-branch GCN model and spectral information. (d) Class features output by SICM-v1 that considers the dual-branch GCN model and spectral-spatial information.

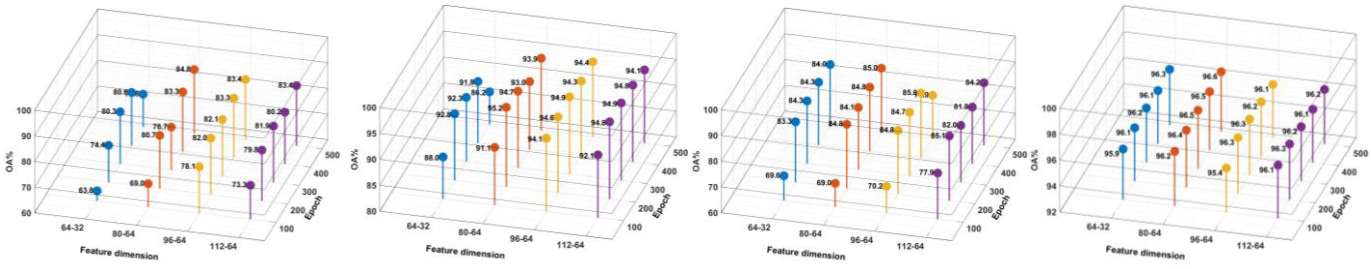
[65]. The visualization results of PCA, SICM-v2, SICM-v6, and SICM-v1 are presented in Figs. 13–16, respectively. The class colors used in these visualizations are consistent with those in the ground truth maps of each dataset shown in Figs. 4–7. The feature distribution obtained by PCA is highly scattered, with considerable overlap among several classes.

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(a) (b) (c) (d)

Fig. 16. t-SNE feature visualization of class features obtained by different comparison methods on the Trento dataset. (a) Class features output by PCA. (b) Class features output by SICM-v2 that considers the ISSG-based GCN and spectral-spatial information. (c) Class features output by SICM-v6 that considers the dual-branch GCN model and spectral information. (d) Class features output by SICM-v1 that considers the dual-branch GCN model and spectral-spatial information.




(a) (b) (c) (d)

Fig. 17. OA results of the proposed method with varied number of features and size of initial spatial window on all datasets. (a) IP. (b) PU. (c) HU. (d) Trento.

In contrast, SICM-v1 achieves better class separability and greater intra-class compactness than the other methods. For example, the distributions of Soybeans-notill (Class 7) and Soybeans-mintill (Class 8) become more compact in the IP dataset, and the samples belonging to Gravel (Class 3) and Bricks (Class 8) are more clearly separated in the PU dataset. Similarly, in the HU dataset, Residential buildings (Class 7), Commercial (Class 8), and Road (Class 9) exhibit improved separability, while in the Trento dataset, Buildings (Class 2) and Roads (Class 6) show more distinct feature distributions. These experimental results further demonstrate that the proposed dual-branch GCN model with contrastive learning can produce more reliable decision boundaries for class-label distributions by effectively reducing inter-class overlap while preserving intra-class compactness.

*2) Parameter Study:* In the parameter sensitivity analysis, we experimentally investigated two key parameters that affect the feature extraction performance of the dual-branch GCN model: 1) feature dimension and the number of training epochs. Fig. 17 presents the OAs of the proposed dual-branch GCN model under different combinations of training epochs and 2) feature dimensions for the graph convolutional and fully connected layers across the different datasets. All other network parameters were kept the same as those used in the previous experiments. The results show that, for a fixed feature dimension, classification accuracy generally improves as the number of training epochs increases. Notably, when the hidden and embedding dimensions of the two GCN layers are set to 96 and 64, respectively, the model achieves performance at 200 epochs that is comparable to that at 300 epochs. This finding indirectly suggests that the proposed method can effectively capture intra-class variations and diversity, thereby facilitating the learning of more robust features while also reducing computational cost to some extent.

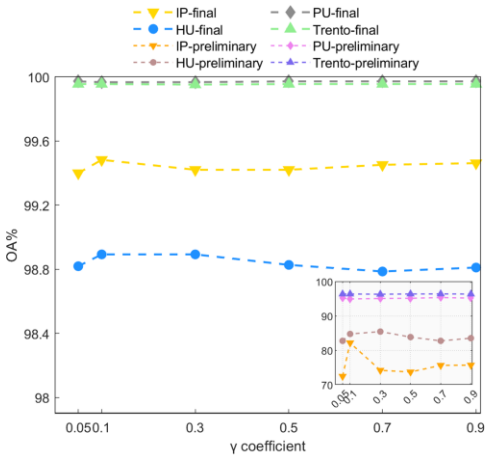



Fig. 18. Sensitivity analysis of the $\gamma$ parameter for ISDG weight function on classification accuracy across four hyperspectral datasets.

Furthermore, we conducted a systematic sensitivity analysis of the parameter $\gamma$ in the ISDG weight function across the four hyperspectral datasets. The parameter $\gamma$ serves as a modulation factor for the weight coefficients of dissimilar samples and prioritizes the most dissimilar intra-class samples according to the spatial-spectral distance ranking, thereby directly influencing the weight allocation $\boldsymbol{A}^{L}_{ISDG}$ defined in (1). Specifically, $\gamma$ regulates the connection weights between sample pairs with different spatial-spectral distances, and thus shapes the topological characteristics of the ISDG structure. Within the dual-branch GCN model, the ISDG branch plays a

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crucial role in modeling intra-class variability, which in turn significantly affects feature extraction and class prediction. As shown in Fig. 18, it can be observed that the OA exhibits only minor fluctuations as $\gamma$ varies on each dataset. When $\gamma$ is small, the weight function becomes more sensitive to spatial-spectral distance differences. This enables the network to assign larger weights to dissimilar intra-class samples and enhances its ability to learn intra-class boundaries. In contrast, when $\gamma$ becomes larger, the weight allocation tends to be more uniform across sample pairs with different spatial-spectral distances, which weakens the effective use of intra-class variability information. Overall, the performance remains relatively stable across a wide range of $\gamma$ values for all datasets, which validates the rationality of the proposed weight function design.

### D. Limitations and Future Work

The core component of the proposed method is a dual-branch GCN model integrated with contrastive learning, which relies on labeled samples during the training phase to construct both the ISSG and ISDG structures. These graph structures are essential for learning robust feature representations. Although the proposed method demonstrates significant effectiveness, it still has certain limitations. In particular, the dual-graph construction strategy depends on the availability of labeled samples for comprehensive intra-class modeling. Specifically, identifying the most dissimilar sample pairs in the ISDG requires a sufficient number of intra-class samples. When the number of labeled samples is extremely limited, data scarcity may lead to sparse graph connections and insufficient representation of intra-class patterns. In future work, we plan to address these limitations by exploring transfer learning and few-shot learning techniques. These strategies may help reduce the dependence on large-scale labeled samples by leveraging prior knowledge or adapting the model to new datasets with only a small number of labeled instances. By incorporating such techniques, we aim to further improve classification performance under limited-label conditions and enhance the applicability and robustness of the proposed method.

## V. Conclusion

In this article, we propose a hyperspectral classification method based on SICM, which fully exploits prior information from multiple perspectives within a given scene. By constructing the ISSG and ISDG, the proposed method jointly models intra-class consistency and variability in labeled samples. On this basis, a dual-branch GCN with contrastive learning was developed to improve feature discrimination and generalization, and a spatial label-refinement strategy was further introduced to exploit local homogeneity in HSIs. Experimental results on several benchmark datasets demonstrate that the proposed method consistently achieves superior classification performance compared with recent graph-based and other state-of-the-art methods. In future work, we will focus on exploring few-shot and transfer learning strategies to reduce the dependence of dual-graph construction on labeled samples.

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